RANGER ELECTRIC VEHICLE (EV) REAR SPRING CAP BOLTS

AFFECTED VEHICLES: ALL 1999 AND CERTAIN 2000 MODEL YEAR RANGER EV

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OVERVIEW

The affected vehicles in this program require inspection of the seam between the rear axle knuckles and the spring caps. The inspection will determine if the spring cap bolts require tightening or if an axle wedge adjustment is necessary. Before any tightening or adjustments are made, tools will be installed on the springs. These tools must remain in place on both springs until all the tightening and adjustments are performed. This procedure provides the details for inspection, tool installation and all necessary tightening or adjustments.

WARNING!

EACH STEP OF THIS PROCEDURE MUST BE READ THOROUGHLY AND PERFORMED COMPLETELY. THE SPRINGS AND SUSPENSION COMPONENTS ARE UNDER EXTREME TENSION. FAILURE TO FOLLOW THIS PROCEDURE EXACTLY MAY RESULT IN DAMAGE TO THE VEHICLE AND/OR SEVERE PERSONAL INJURY.

INSPECTION AND TOOL INSTALLATION

NOTE: Two (2) spring adjusting tools (204-F002) are required to properly perform this procedure.

1. NOTE: Some hoists will require the use of taller hoist adapters, to provide clearance for the spring adjuster tools that will be installed.

   Raise and support the vehicle.

2. Remove both rear wheel and tire assemblies.

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3. Thoroughly clean the seam between the spring cap and the axle knuckle on both the right and left springs. Inspect the seams for any visible gaps. See Figure 1.
   - All spring(s) that have a visible gap(s) at the seams require an axle wedge adjustment. To prepare for this adjustment, the wedge adjusting bolt must be removed at this time. This bolt cannot be removed once the spring adjusting tools are installed.
   - All spring(s) that do not have a visible gap at the seams require only tightening the spring cap bolts after the spring adjusting tools are installed.

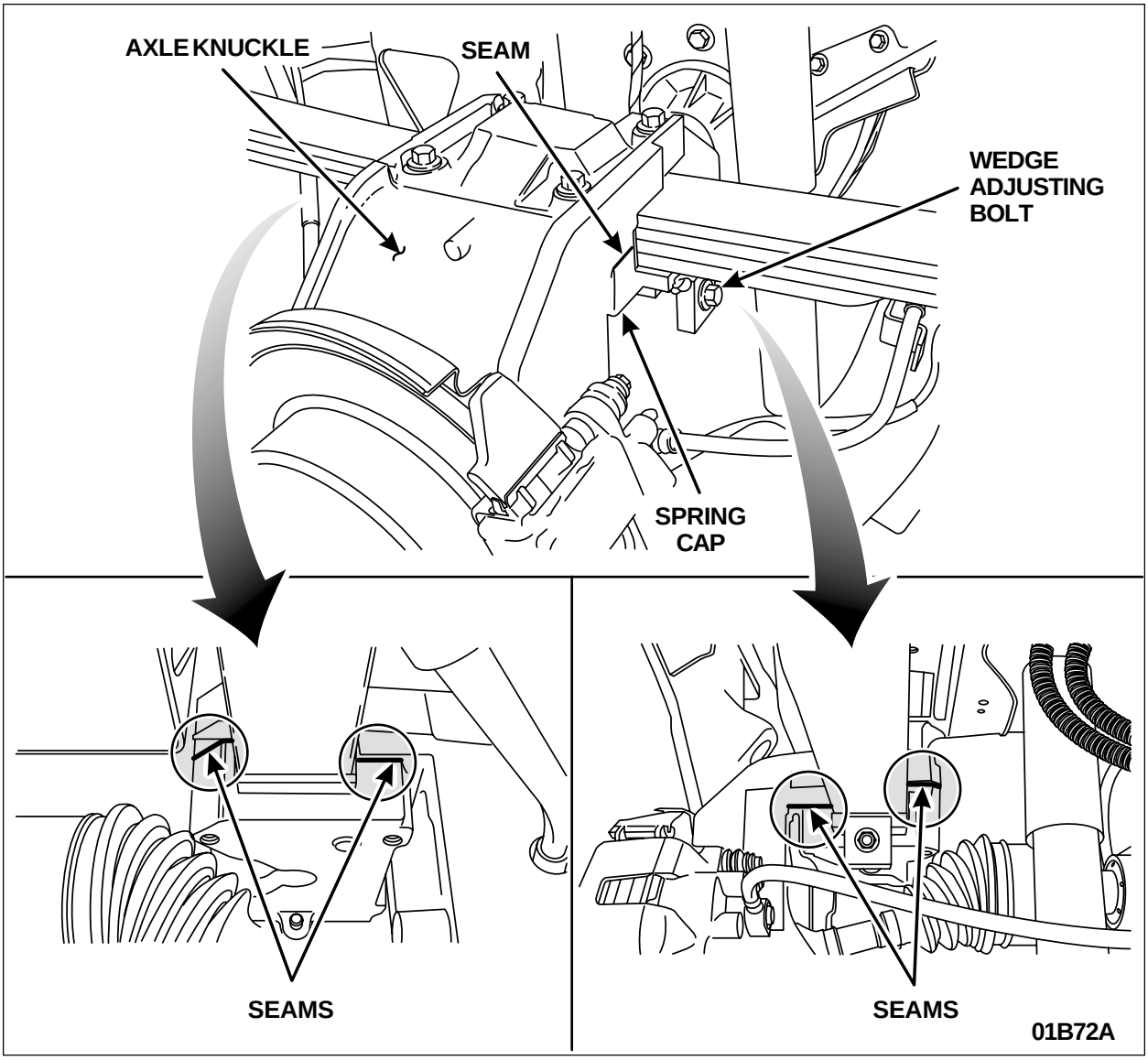


FIGURE 1

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4. NOTE: The spring adjustment tools must be installed on both sides of the vehicle and must remain in place during the spring cap tightening/axle wedge adjustment.

   CAUTION: The spring adjusting bar must be installed from the front of the axle and UNDER the rear brake hose to avoid damaging the hose or the halfshaft boot.

   With the aid of an assistant, install spring adjusting tools (204-F002) to the left and right rear spring as follows. See Figures 2, 3 and 4.

   - Verify that the spring adjusting bars are assembled as illustrated. The stop blocks and adjuster pads MUST be properly positioned before installing the bars.
   - Position the spring adjusting bars below and parallel to the rear springs.
   - Install the spring adjusting clamps with the jaws of the clamps facing outboard. Hand tighten the clamps.
   - Position the adjusting bar until the stop block squarely contacts the spring cap. Adjust the stop block until there is 50 mm (2-inches) between the inner face of the tool and the spring cap. See the detailed illustration in Figure 3.
   - With hand tools only, alternately tighten the spring adjusting clamps until the clamp stop sleeves contact the clamp frames. Due to differences in springs, the clamping force applied to the springs will vary between vehicles. On some springs, the adjusting bar may not fully contact the spring with the full clamping force applied.

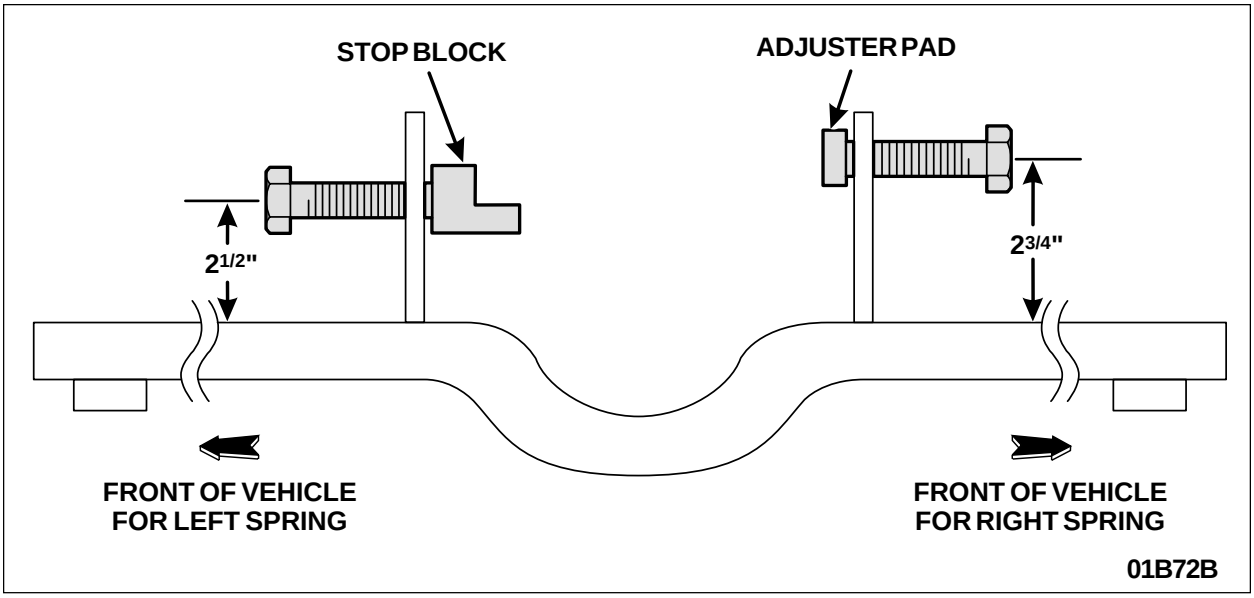


FIGURE 2

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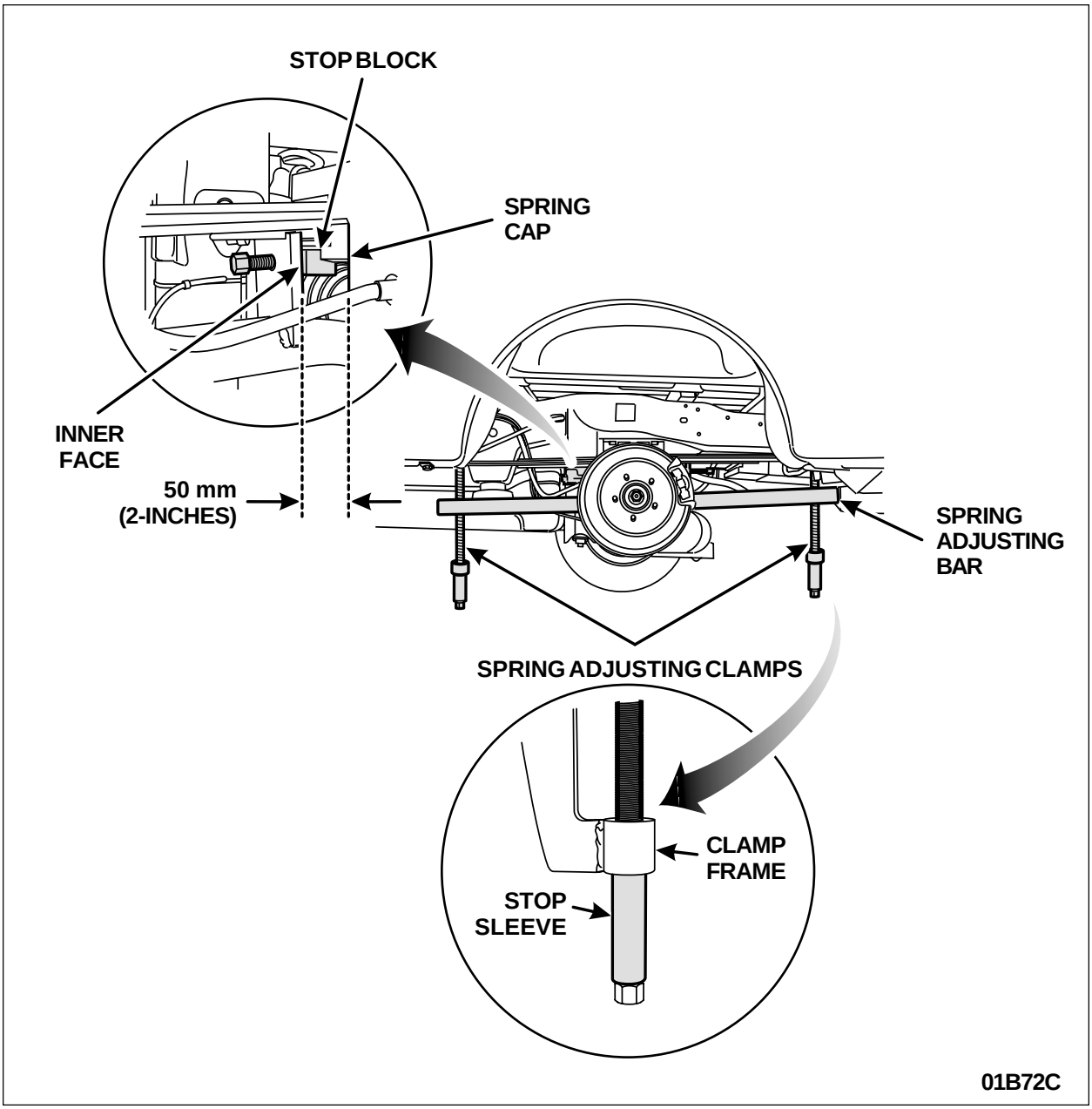


FIGURE 3

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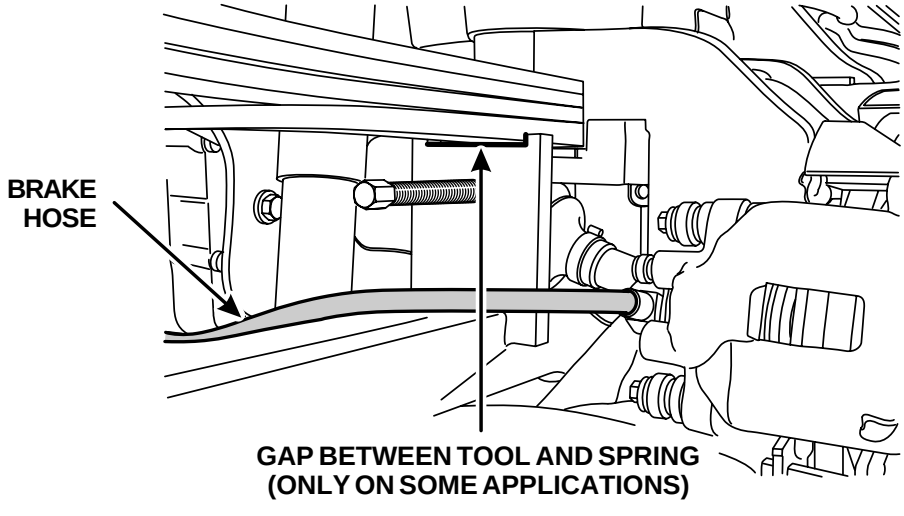


FIGURE 4

5. **WARNING: THE SHOCK ABSORBERS SHOULD ONLY BE DISCONNECTED WITH THE AXLE SUPPORTED BY SHORT SAFETY STANDS WHILE THE FRONT WHEELS ARE MAKING CONTACT WITH THE FLOOR. ATTEMPTING TO DISCONNECT THE SHOCK ABSORBER WHILE THE VEHICLE IS SUPPORTED BY THE HOIST MAY RESULT IN SUDDEN SHIFTING OF THE VEHICLE ON THE HOIST, WHICH CAN CAUSE PERSONAL INJURY.**

   CAUTION: Attempting to disconnect the shock absorbers without supporting the axle or without the spring adjusting tools installed will cause damage to the axle halfshafts.

   Position two (2) short safety stands directly under the rear axle. Lower the hoist so the rear axle is supported by the stands and the front wheels fully contact the floor. See Figure 5.

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6. Remove the lower mounting bolts from the left and right rear shock absorbers. See Figure 5.

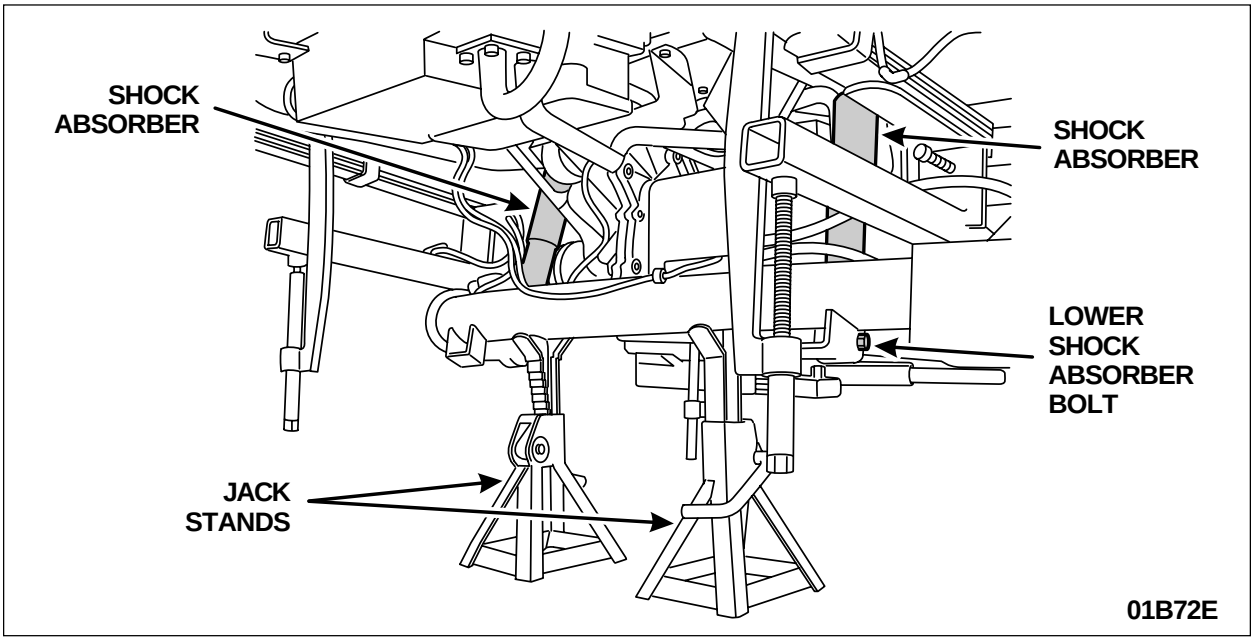


FIGURE 5

7. **CAUTION: Once the shock absorbers have been disconnected and the vehicle raised, do not remove or loosen the spring adjusting tools for ANY reason. If either tool is loosened or removed, the axle halfshafts will be damaged. If the tool needs to be repositioned, the vehicle must be lowered and the axle must be supported.**

   Verify that the adjusting bar clamp screws or the stop blocks have not shifted, then raise the vehicle.

8. The spring adjusting tools have now been properly installed and must remain in place throughout the Spring Cap Tightening/Axle Wedge Adjustment.

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SPRING CAP TIGHTENING/AXLE WEDGE ADJUSTMENT

NOTE: The spring adjustment tools must remain in place on both springs while performing any part of this procedure.

SPRING CAP TIGHTENING

NOTE: Perform this procedure only on the spring(s) that do not have a visible gap(s) between the spring cap and the axle knuckle.

1. Tighten the spring cap bolts to 103 Nm (76 lb-ft). See Figure 6.

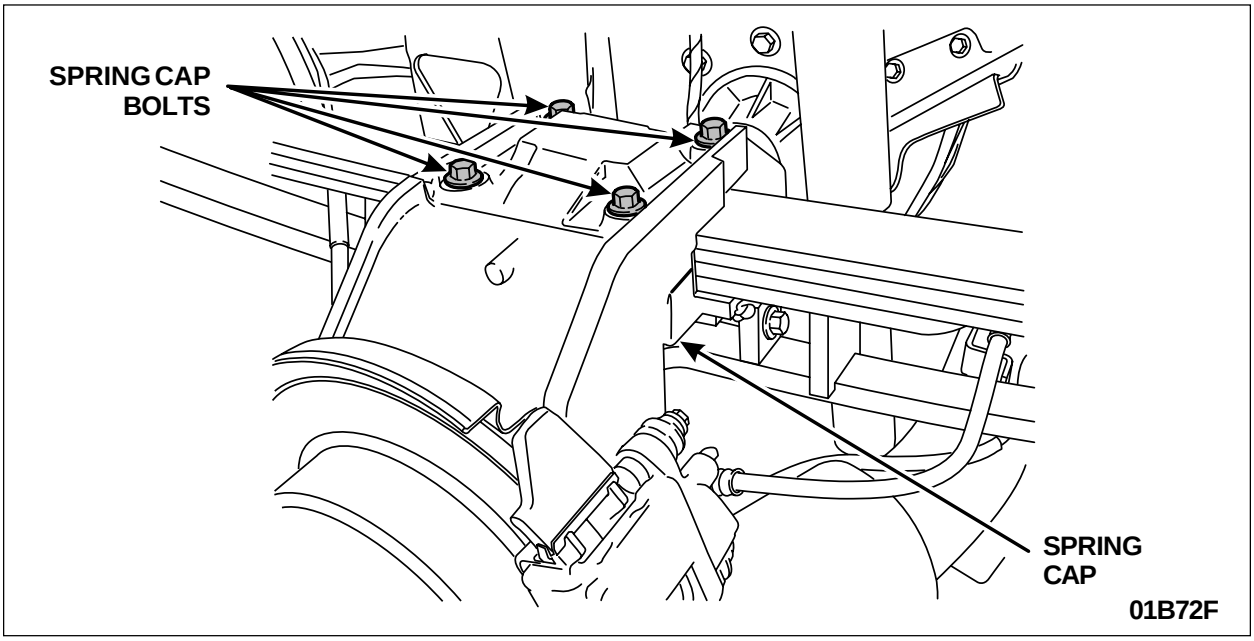


FIGURE 6

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AXLE WEDGE ADJUSTMENT

NOTE: Perform this procedure only on the spring(s) that have a visible gap(s) between the spring cap and the axle knuckle.

1. Loosen all four (4) spring cap bolts by two (2) complete revolutions. See Figure 7.

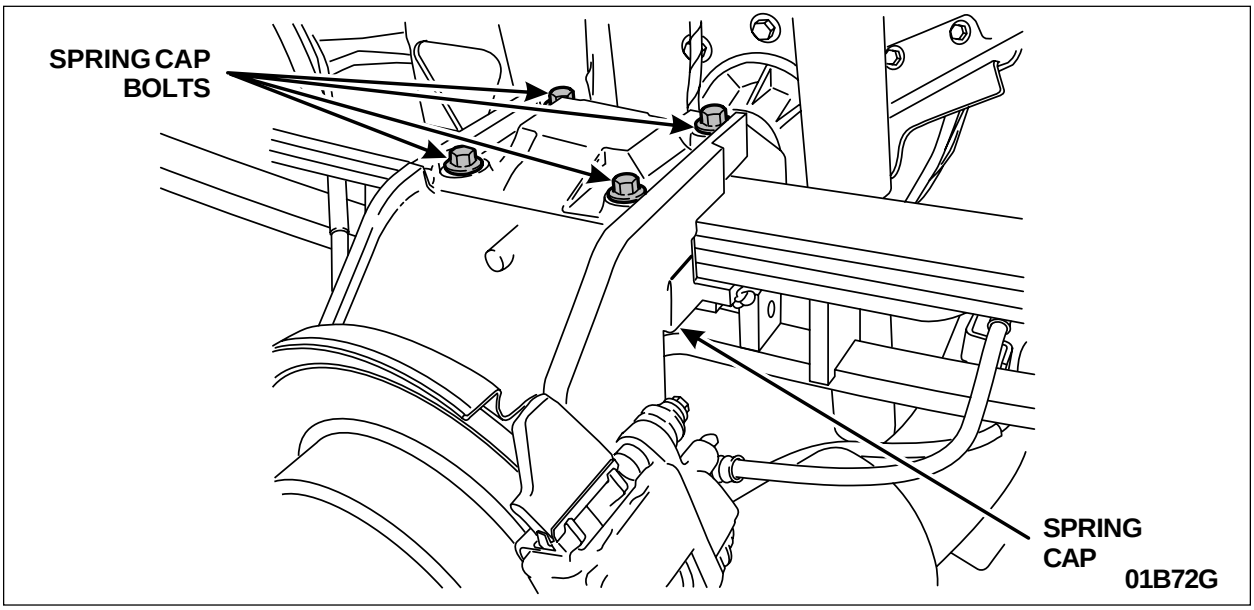


FIGURE 7

2. **CAUTION: Only use a wood block to drive the aluminum axle wedge. Using a metal tool will damage the wedge.**

   Using a hammer and a wood block, drive the tapered end of the axle wedge until it is flush with the spring cap. See Figure 8.

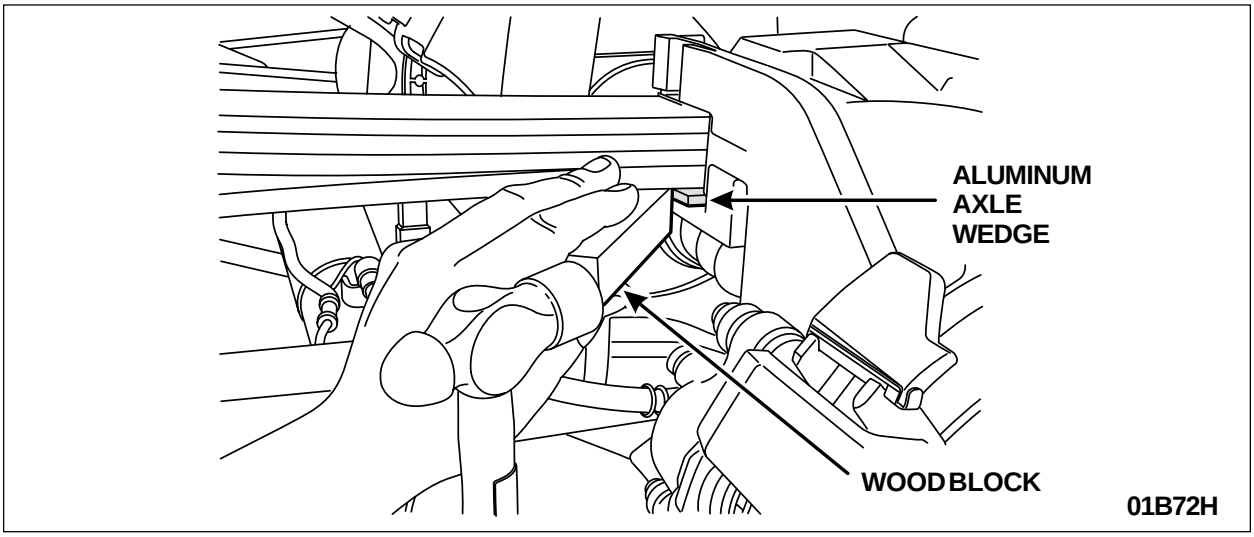


FIGURE 8

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3. Clean any debris from between the spring cap and axle knuckle mating surfaces with compressed air.
4. Tighten the four (4) spring cap bolts to 103 Nm (76 lb-ft).
5. Verify that the stop block squarely contacts the spring cap. DO NOT TIGHTEN. See Figure 9.
6. Position the wedge adjuster so it squarely contacts the axle wedge. See Figure 9.

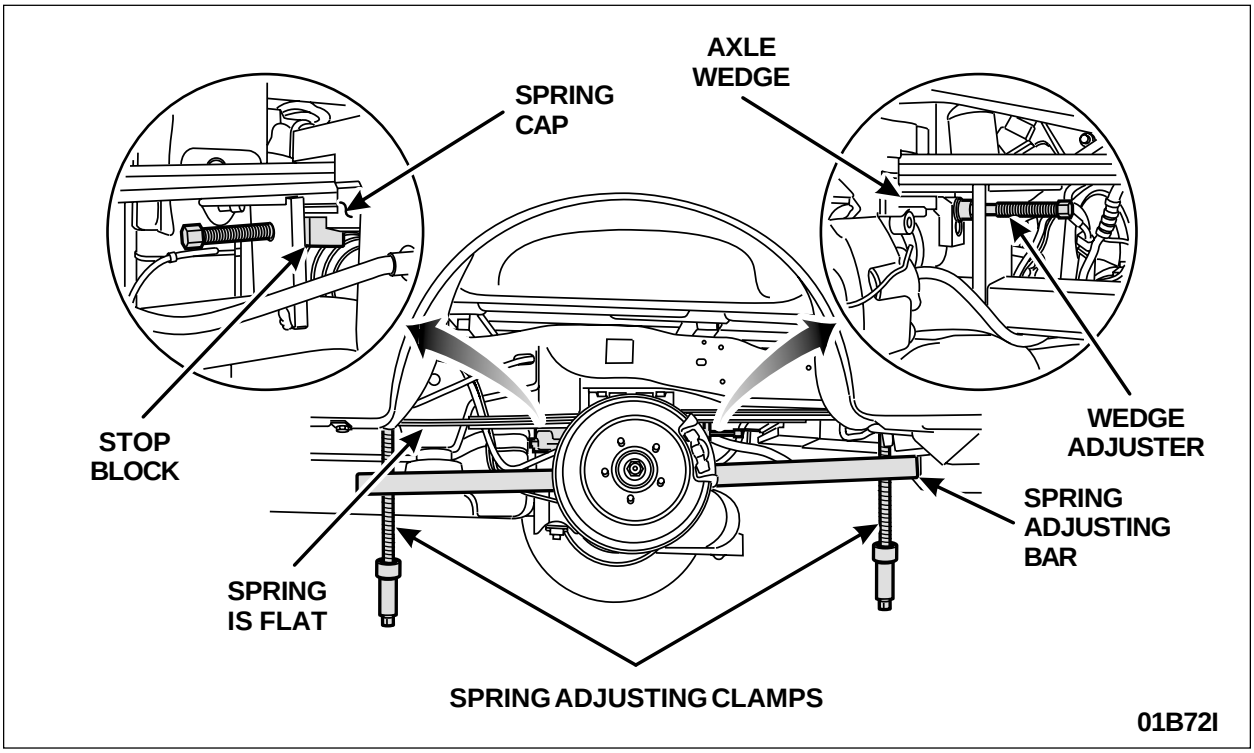


FIGURE 9

7. Press the axle wedge into place by tightening the wedge adjuster to 60 Nm (44 lb-ft).
8. Back out the wedge adjuster and stop block.

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TOOL REMOVAL

NOTE: Do not perform the tool removal until both springs have had either the spring caps tightened or the axle wedge adjustment performed.

1. Position two (2) short safety stands directly under the rear axle. Lower the hoist so the rear axle is supported by the stands and the front wheels fully contact the floor.
2. Install the lower mounting bolts to the left and right rear shock absorbers. Tighten the bolts to 55 Nm (41 lb-ft).
3. Raise the vehicle and remove the safety stands.
4. With the aid of an assistant, remove the spring adjustment tools from both springs.
5. If removed, apply Ford Threadlock and Sealer (EOAZ-19554-AA) or equivalent to the wedge adjuster bolt(s). Install the wedge adjuster bolt(s) and tighten to 5 Nm (44 lb-in). See Figure 10.

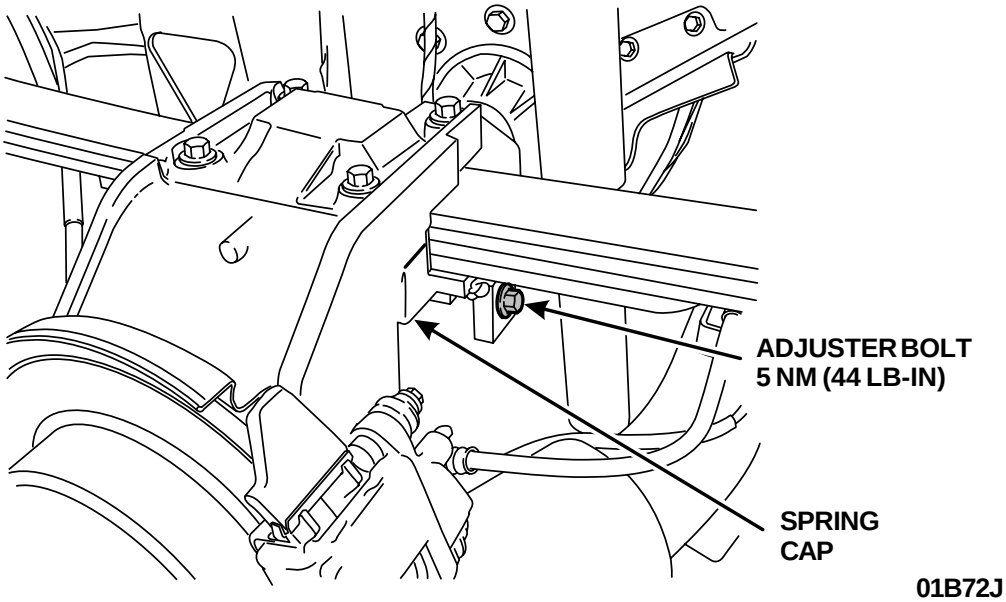


FIGURE 10

6. Install the wheel and tire assemblies. Tighten to 115 Nm (85 lb-ft).
7. Lower the vehicle.

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